

Toy Example: Confusion and Diffusion (12-Bit Key, 16-Bit Block)

Plaintext Block (16 bits)

P = 1010 1100 0011 0110

This represents structured data, such as a formatted record or file header.

Encryption Key (12 bits)

K = 1101 0110 1011

The key will influence encryption but will not, by itself, provide security.

CONFUSION (Hiding the Key Relationship)

Step 1: Expand the Key

Because the plaintext block is 16 bits, we first expand the 12-bit key using a toy rule:

Repeat the first 4 bits of the key.

K = 1101 0110 1011

Expanded = **1101 0110 1011 1101**

This step only aligns sizes. It does not add security.

Step 2: Key Mixing (XOR)

We now combine the plaintext with the expanded key:

P = 1010 1100 0011 0110

K = 1101 0110 1011 1101

X = 0111 1010 1000 1011

At this point:

- The key affects every bit
- The relationship is still linear
- An attacker could still reason algebraically

This step prepares for confusion but does not yet provide it.

Confusion (achieved through substitution)

We now apply a nonlinear substitution.

Rule:

- Split the data into 4-bit chunks

- Replace each chunk using a fixed lookup table, where there is a 4-bit code for every 4-bit combination.

Apply the substitution:

X = 0111 | 1010 | 1000 | 1011
 ↓ ↓ ↓ ↓
1000 | 0110 | 0011 | 1100

Result after substitution:

C = 1000 0110 0011 1100

What Confusion Achieves

- The relationship between the key and the output is now nonlinear
- The key cannot be recovered using simple equations
- Small key changes cause unpredictable output changes

Patterns in the key relationship are hidden, but plaintext structure may still exist because

- The values have been scrambled in a way that hides how the key affects the output.
- However, each 4-bit chunk stayed in the same position.
- Local layout and structure may still be visible.

DIFFUSION (Destroying Plaintext Structure)

Step 4: Diffusion (achieved via permutation)

We now rearrange bits to spread influence across the block.

Rule:

Swap the left and right 8-bit halves.

Before:

1000 0110 | 0011 1100

After:

0011 1100 | 1000 0110

Result:

D = 0011 1100 1000 0110

What Diffusion Achieves

- Bits that were originally grouped together are now separated
- Plaintext structure becomes harder to recognize
- Changes will spread further if this process is repeated

Diffusion hides where information came from, not which key was used.

What Happens If One Plaintext Bit Changes?

Original plaintext:

1010 1100 0011 0110

Flip one bit:

1010 1100 0011 0111

After confusion and diffusion:

- Multiple output bits change
- No predictable pattern remains
- This is the beginning of the avalanche effect

This is a very simple example because

- Uses only one substitution step
- Uses only one permutation step
- Uses no rounds

Real block ciphers:

- Repeat confusion and diffusion many times
- Use stronger substitutions
- Achieve full avalanche through rounds

Summary: *Confusion hides how the key affects the output. Diffusion hides how the plaintext is structured. Repeating both makes encryption secure.*